Linear Mixed Models

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Warmup

Work on the warmup activity (handout).

Warmup

$\beta_0 + u_i$ is intercept for neighborhood i

Data on prices and other information for 1561 Airbnb rentals in Chicago. Our goal is to model the relationship between price and overall satisfaction score.

- `price`: The price in US dollars of a one night rental
- `overallSatisfaction`: an average satisfaction score, given as numbers between 2.5 and 5
- `neighborhood`: the neighborhood in which the rental is located. There are 43 neighborhoods, and 25 - 50 rentals per neighborhood

Question: What would an appropriate linear mixed model for this data and research question look like?

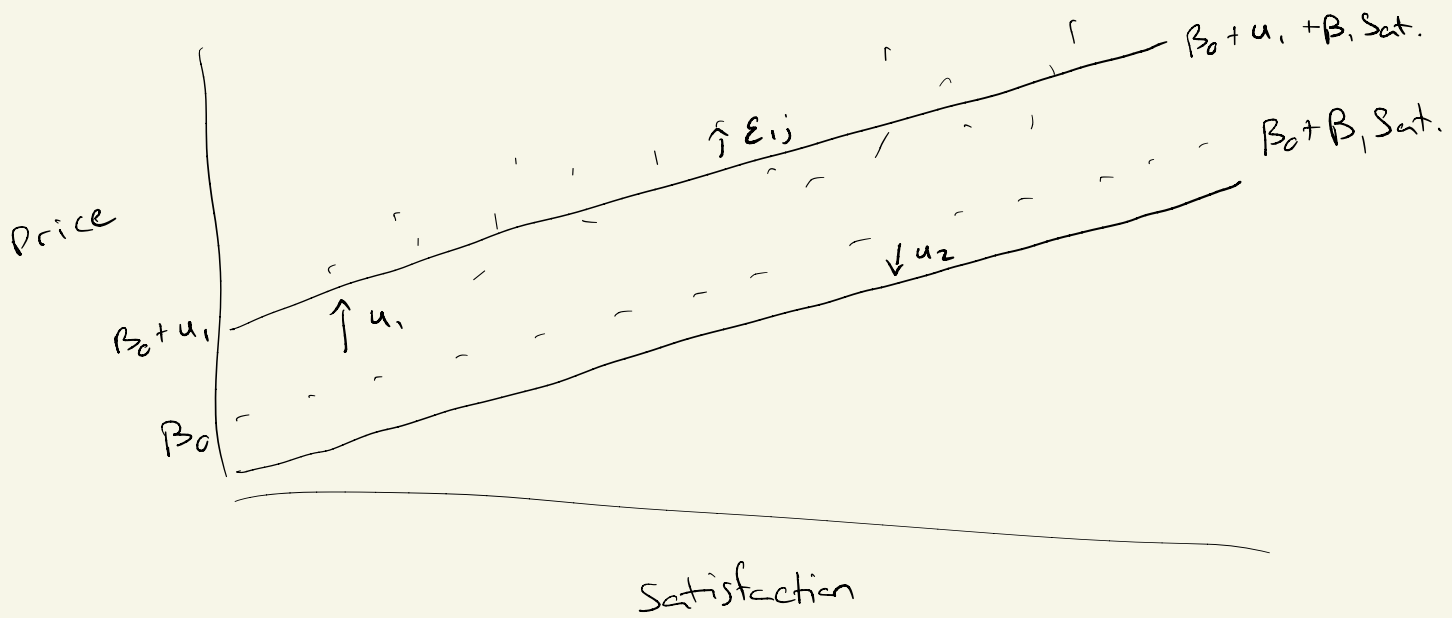
$$\text{Price}_{ij} = \beta_0 + u_i + \beta_1 \text{Satisfaction}_{ij} + \varepsilon_{ij}$$

i = neighborhood

j = rental w/in neighborhood

$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$

$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$


$\beta_0 + u_1 + \beta_1 Sat.$
$\beta_0 + \beta_1 Sat.$
$\uparrow \varepsilon_{ij}$
Price
$\beta_0 + u_1$
$\uparrow u_1$
β_0
$\downarrow u_2$
Satisfaction

Warmup

$i = 1, \ldots, m$

$j = 1, \ldots, n_i$

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Random effects:
 Groups       Name        Variance Std.Dev.
 neighborhood (Intercept) 1048     32.37
 Residual                 6762     82.23
Number of obs: 1561, groups: neighborhood, 43
Fixed effects:
                      Estimate Std. Error t value
(Intercept)             27.275     30.547   0.893
overall_satisfaction    14.809      6.303   2.350
```

$\hat{\sigma}_u^2$ (Variance for neighborhood)

$\hat{\sigma}_\varepsilon^2$ (Residual variance)

m = 43 neighborhoods

$n = \sum_{i=1}^{m} n_i$ = 1561 observations

$\hat{\beta}_0$ = 27.275 (Intercept)

$\hat{\beta}_1$ = 14.809 (overall_satisfaction)

no p-vals here!

$E[\text{price}_{ij} \mid \text{satisfaction}_{ij} = 0, u_i] = \beta_0 + u_i$ (intercept for neighborhood i)

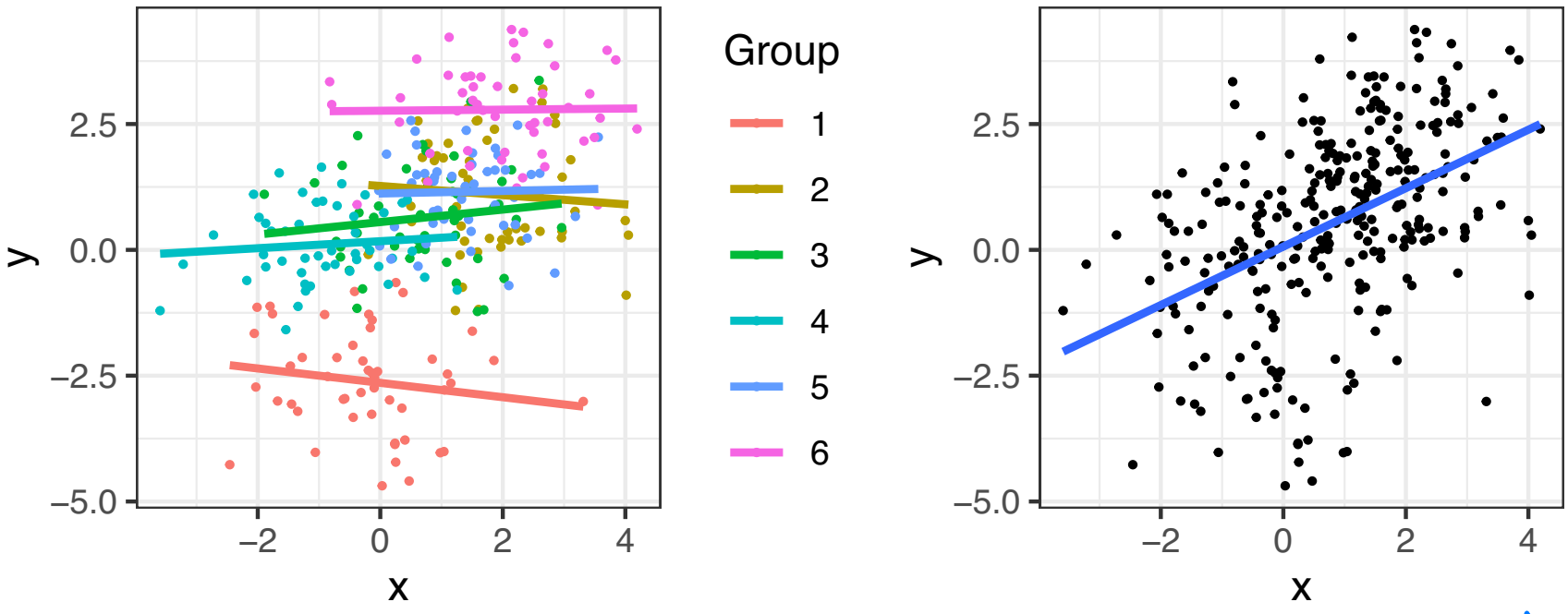
$E[E[\text{price}_{ij} \mid \text{satisfaction}_{ij} = 0, u_i]] = \beta_0$ (average across neighborhoods)

$\hat{\beta}_1 = 14.81$

For a fixed neighborhood, we estimate a one-unit increase in satisfaction is associated with an average increase of $14.81 in price

Last time: class activity

In simulation, $\beta_1 = 0$



Linear regression: $\widehat{\beta}_1 = 0.5806,\ \widehat{SE}(\widehat{\beta}_1) = 0.066$ ← confidently wrong!

LMM: $\widehat{\beta}_1 = 0.00207,\ \widehat{SE}(\widehat{\beta}_1) = 0.0546$

↑ much closer to truth

Issues with ignoring group structure

- Estimating the wrong coefficients
- Standard errors are too small
- Inflated type I error rates for hypothesis tests

Models with a random intercept

observe data (x_{ij}, y_{ij}) from m groups $(i = 1, \dots, m)$ with n_i observations per group $(j = 1, \dots, n_i)$, and $n = \sum_{i=1}^{m} n_i$

$$y_{ij} = u_i + x_{ij}^T \beta + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

$$\begin{bmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{21} \\ y_{22} \\ \vdots \\ \cdot \end{bmatrix} = \begin{bmatrix} x_{11}^T \\ x_{12}^T \\ \cdot \\ \cdot \\ \cdot \\ x_{mn_m}^T \end{bmatrix} \beta + \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \cdots & & & & \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ \cdot \\ \cdot \\ \cdot \\ u_m \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \cdot \\ \cdot \\ \cdot \\ \varepsilon_{mn_m} \end{bmatrix}$$

$$y = X\beta + Zu + \varepsilon$$

Class activity

Work on the class activity (handout).